Analysing 2025 MLB Postseason Statcast data by Quantitative view

Introduction

While the advantage of fastballs in baseball is pretty much known by everyone, this project e xplores the relationship between pitching characteristics and hits through data analysis using S tatcast data from the 2025 MLB postseason. In particular, this project views pitching as a pro babilistic event and interprets hits as a risk event. This approach is similar to the way financi al markets evaluate the expected return and risk of assets, and the physical characteristics of pitching are analysed to observe whether this holds true in actual game data.

Data Curation

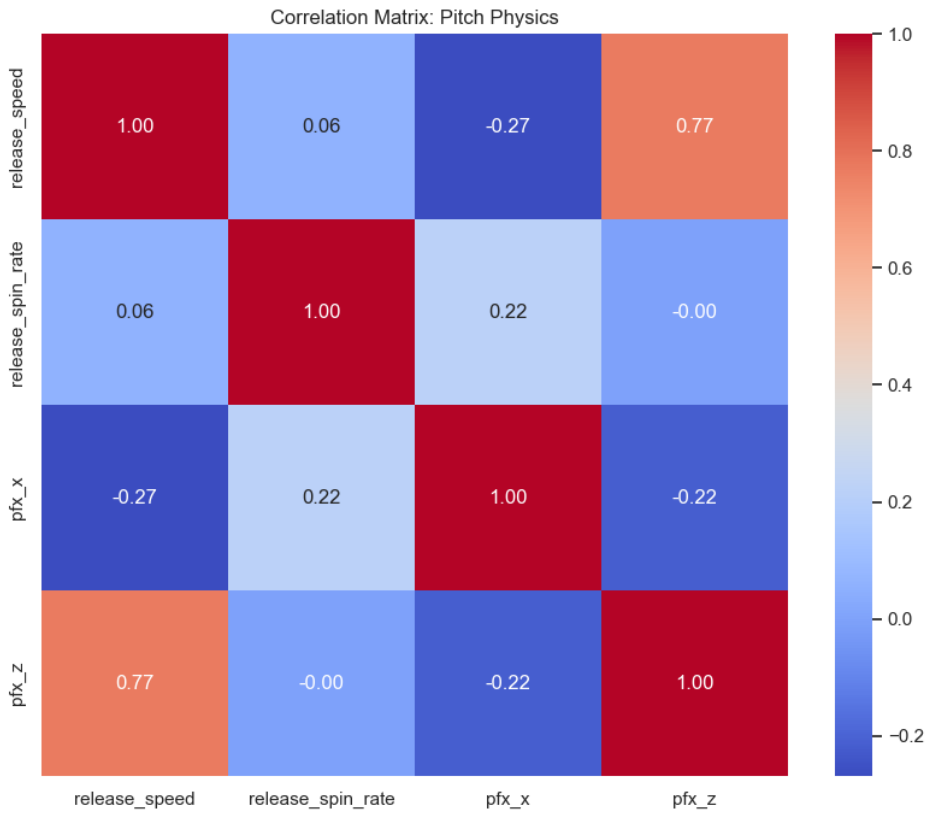
This dataset contains velocity, spin rate, displacement, strike zone location, and game outcome (event) data for each pitch. This paper describes the technical decision-making process for i mproving the original dataset, which consists of over 14,000 elements, focusing on maintainin g data integrity, a key element of quantitative analysis.

Feature Engineering: Due to the lack of direct indicators for heat occurrence in raw data, the 'event' column was used to define the heat occurrence as a binary variable. Instances categorised as single, double, triple, and home run were coded as 1 (hit), and all other results were coded as 0 (out). In this preparatory step, we set a baseline for all subsequent analyses.

Data cleaning: The 'release_spin_rate' measurement was found to be missing less than 0.1% due to a sensor error, so the row was removed to maintain data quality.

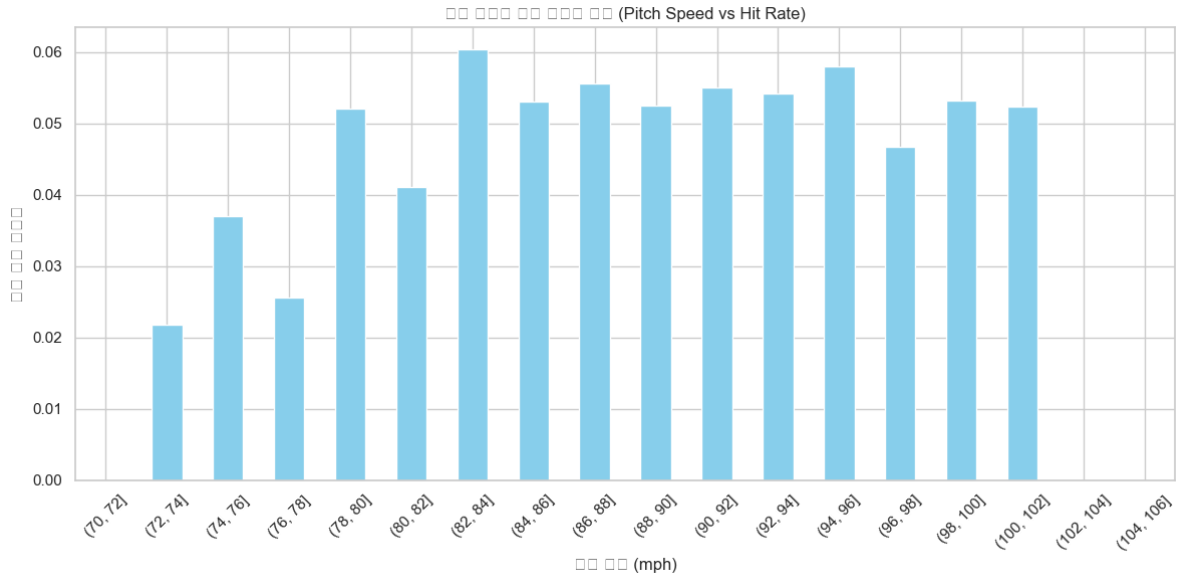
Key Findings

Figure 1: Correlation Heatmap Analysis



A correlation matrix analysis revealed a high correlation of 0.77 between velocity (release_speed) and vertical change (pfx_z). This suggests that fastballs not only arrive quickly but also give the illusion of 'rising' over the batter's bat due to their trajectory and speed. However, the correlation between velocity and spin rate was very low at 0.06, confirming that the two metrics are independent variables. This suggests that evaluating pitching should consider several physical factors together, not just one.

Figure 2: Regression Hit Analysis



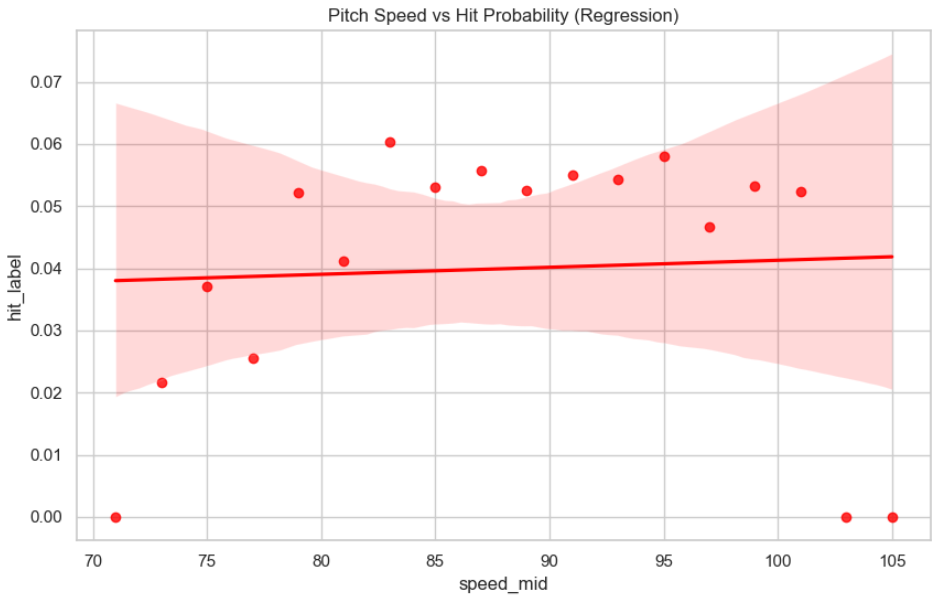
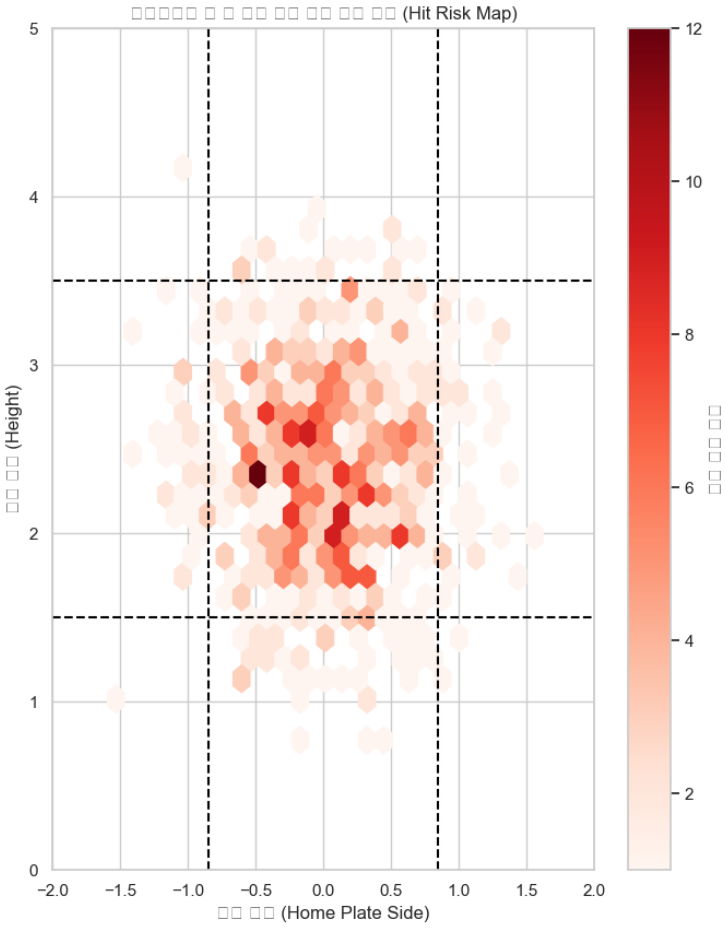


Figure 2 presents the relationship between velocity and hits using both a bar chart and a regr ession plot. The first bar chart shows the observed hit rate in each 2-mph interval, intuitively showing the expected value of hits according to the speed. The regression plot adds a linear regression line above the average hit rate for each interval and displays confidence intervals ("pink shaded intervals"), allowing for observation of both the overall trend and the variability of results as the speed increases. The confidence intervals narrow as the speed increases, whi ch can be called the "predictability zone."

In summary, the faster the pitch interval, the more predictive the model becomes, meaning th e less likely it is to allow unexpected hits (outliers). Furthermore, the analysis results show a noticeable decrease in hit probability starting around 96 mph, which can be interpreted as a r ange where it becomes more difficult for batters to accurately contact the ball at speeds abov e that level. This pattern is always shown in both the bar graph and the regression analysis c onfidence intervals.

Figure 3: Spatial Risk Density



Using kernel density estimation techniques to analyse the density of batted balls within the st rike zone, the red area shows a concentration of balls in the centre of the strike zone. This i ndicates that, regardless of speed, the centre of the strike zone (plate_x = 0, plate_z = 2.5) r epresents the "Maximum Risk Area," where the hit probability density is highest. Pitches thro wn at the edges of the strike zone had significantly lower expected batting averages, which r eally shows how much it matters strategically to place the ball precisely when trying to contr ol batted ball risk.

Conclusion

This study analysed 2025 MLB Postseason Statcast data to interpret pitching physical characte ristics using probability and expected value, and to examine their relationship with hit risk. T he data confirm that hit probability decreases beyond a specific velocity threshold and identif y the centre of the strike zone as a relatively high-risk area. While this project is focused on identifying patterns rather than establishing clear causal relationships, it shows that quantitativ e analysis systems commonly used in finance can be successfully applied to sports data. Futu re research can be expand this analysis through advanced statistical analysis or predictive mod elling.